

The first E-STATCOM in operation: Mehrum Nord

TenneT TSO TenneT Germany, 2025



Challenge

- **First** of its kind
- Demanding time schedule

As conventional power plants are phased out, the grid is losing the natural **stabilizing forces** that once kept frequency stable. Maintaining resilience in this new energy landscape demanded a **first-of-its-kind solution worldwide**, delivered under a **highly demanding time schedule**. With no blueprint to follow, innovation, speed, and reliability had to come together seamlessly.

Technical solution

At the Mehrum site, a new generation of grid technology was realized with the deployment of the world's first E-STATCOM. The solution is based on SVC PLUS® technology, rated at **±300 Mvar** and combined with a **200 MW ultracapacitor storage unit**. This configuration enables the system to actively inject high levels of active power into the grid within milliseconds, closely emulating the dynamic behavior of conventional power plants.

By providing synthetic system inertia alongside fast and precise voltage control, the E-STATCOM delivers a compact and highly integrated approach to **grid stabilization**. Its modular design and short response time allow it to react instantly to frequency deviations, forming a key technical foundation for stable grid operation in power systems with a **growing share of renewables**.

“The energy transition means moving from conventional power plants with heavy rotating machines to inverter-based sources like wind and solar”

German Kuhn, Siemens Energy



Contribution to energy transition

- **Enables large-scale renewable integration** by providing the stability needed to absorb growing wind output and manage fluctuating power flows.
- **Replaces stability lost through coal and nuclear phase-out** with synthetic system inertia and essential grid services traditionally delivered by conventional plants.

- **Strengthens Germany’s north–south transmission backbone**, supporting reliable transfer from northern generation zones to southern demand hubs.

Together, these contributions make Mehrum Nord a key enabler of Germany’s energy transition. Realized in **partnership with TenneT**, the project shows how advanced grid technology and strong collaboration can secure a resilient, decarbonized power system.

Benefits

The Mehrum E-STATCOM project sets a new standard for how modern power systems can be stabilized in an increasingly renewable-driven environment. By combining SVC PLUS® technology with a **high-power ultracapacitor storage unit**, the system delivers a level of dynamic grid support that was previously only achievable with conventional power plants. It provides effective system inertia by injecting high active power precisely when frequency deviations occur, enabling a fast and controlled response to grid disturbances.

With its **millisecond response time**, the solution significantly enhances grid robustness and plays a critical role in preventing cascading failures and blackouts. At the same time, the compact and highly integrated design allows this advanced functionality to be realized in a **cost-efficient footprint**, making it an attractive alternative to traditional stabilization assets. The result is a future-ready solution that strengthens frequency stability, supports secure grid operation, and creates a reliable technical foundation for the continued expansion of renewable energy.



Mehrum Nord, inside view

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